

제어 장벽 함수 기반 6-자유도 자기부상 시스템의 안전 제어

Control Barrier Function-Based Safety-Critical Control of 6-DOF Magnetic Levitation System

*김창환(연세대학교), #윤준영(연세대학교)
*C. H. Kim, #J. Y. Yoon

Key words : Control barrier function, Safety-critical control, 6-DOF magnetic levitation

This work presents a control barrier function (CBF)-based approach for safety-critical control of a six-degree-of-freedom (6-DOF) magnetic levitation system. Conventional motion controllers, such as PID and LQR, have been widely employed owing to their simple structure and low implementation complexity, but they do not account for position constraints and may therefore lead to collisions with surrounding structures. MPC can address this limitation by incorporating constraints into the control formulation, but its computational burden limits the achievable control rate, making it difficult to attain the 10 kHz sampling frequency required for high-precision control. To address this issue, a CBF-based safety filter is introduced to enforce position constraints while preserving the high control rate of the underlying controller. A linear high-order control barrier function (HOCBF) is designed based on the electromagnetic and dynamic models of the system and implemented to ensure that the mover's position remains within a predefined safe region. Experimental results demonstrate that the proposed approach successfully enforces position constraints without compromising the 10 kHz sampling rate, thereby improving safety while maintaining high-precision control performance. These results indicate that the proposed CBF-based control is suitable for safety-critical mechatronic systems requiring fast sampling rates for precise motion control.

후기 이 논문은 정부(과학기술정보통신부)의 재원으로 한국연구재단의 지원(No. RS-2023-00208437)을 받아 연구되었음.

*발표자, #교신저자(junyoung.yoon@yonsei.ac.kr)